

Efficient In-Context Learning for 3D Medical Image Segmentation

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by Tidiane Camaret Ndir



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Efficient In-Context Learning for 3D Medical Image Segmentation

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Declaration

I hereby declare, that I am the sole author and composer of my thesis and that no other sources or learning aids, other than those listed, have been used.

Furthermore, I declare that I have acknowledged the work of others by providing detailed references of said work.

I also hereby declare that my thesis has not been prepared for another examination or assignment, either in its entirety or excerpts thereof.

Freiburg, September 27, 2026

Place, date

Tidiane Camaret Ndir

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Abstract

Zusammenfassung

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1 Motivation

In-context learning (ICL) recasts segmentation as few-shot matching: given a target scan and a small set of (image, mask) context pairs illustrating the structure of interest, a model predicts a mask for the target with no task-specific fine-tuning. This removes the need to retrain for every new anatomical class – valuable for structures that are rare, newly defined, or annotated at only a handful of sites.

Most ICL segmentation models were designed and validated in 2D. Moving to native 3D volumes is attractive – CT/MRI are inherently volumetric, and slice-wise processing discards through-plane context needed to resolve thin or elongated structures – but it sharpens an already difficult resolution/compute trade-off: dense cross-attention between a target volume and its context volumes scales with the cube of side length, so a design affordable at 64^3 becomes impractical at 256^3 .

This trade-off is not one problem but three, and this thesis addresses each with its own mechanism:

1. **How should image and label evidence be combined?** Concatenating image and mask channels before a shared encoder, or fusing them once through a single operation, commits to one fusion point before most of the network’s capacity is applied. section 3 proposes *dual attention*, keeping the two streams separate and letting them exchange information repeatedly over a modality axis and a position axis.
2. **Where should compute be spent?** Re-scanning an entire volume with a sliding window at every processing step wastes compute on regions already known not to contain the structure of interest. section 3 proposes a *coarse-to-fine cascade* that restricts fine processing to the region a coarse pass has already localized, connected across levels by a query prior and register tokens that carry a gradient path between resolutions.
3. **Where does training signal come from beyond limited annotation?** Real (image, mask) pairs are limited to the classes and appearance distribution covered by the training cohort. section 3 proposes a *synthetic task generator* that repaints real anatomical geometry with novel, controllable appearance, producing label-perfect training tasks beyond the annotated cohort.

Contributions.

- A dual attention mechanism for in-context 3D segmentation, contrasted against early-concatenation (Medverse [8]) and one-shot fusion (Iris [6]) baselines.
- A coarse-to-fine cascade in which the query prior, region-restricted fine processing, and cross-level register training are evaluated against the sliding-window-over-the-whole-volume design used by prior work.
- A GMM-based synthetic task generator with a texture-noise variant that better matches real tissue’s spatial autocorrelation, evaluated against real-only and i.i.d.-noise training.
- Evaluation on TotalSegmentator CT and a range of out-of-distribution cohorts and modalities, reporting where each mechanism helps and where it does not.

Outline. section 2 introduces in-context segmentation, the volumetric encoder and attention machinery the method builds on, and synthetic-data background, then positions UniverSeg, Iris, and Medverse against the three axes above. section 3 details the model. section 4 evaluates each axis in turn. section 5 synthesizes what the evidence supports and what remains open, and section 6 summarizes.

2 Fundamentals

2.1 In-Context Segmentation

Given a target image I^t and a context set $\{(I_k^c, L_k^c)\}_{k=1}^K$ of image/mask pairs for the same class, in-context segmentation predicts a mask $\hat{L}^t$ for the target in a single forward pass, with no gradient update and no class-specific parameters. This sits between two more familiar paradigms: closed-set supervised segmentation, which fixes a label set at training time and must be retrained (or fine-tuned) to add a class; and interactive segmentation, where a human supplies clicks or boxes on the target image itself, per image, at inference time. In-context segmentation instead supplies the “prompt” as a handful of *other*, already-labeled examples of the same class – closer to how a radiologist might be shown a few example cases of a rare finding and asked to find it in a new scan. This is valuable in medical imaging specifically because annotation requires domain expertise, new structures and pathologies are defined faster than segmentation datasets can be built for them, and per-site protocol differences limit how far a closed-set model trained elsewhere transfers.

2.2 Volumetric Encoders and Attention

Convolutional volumetric backbones. 3D U-Net [4] and its self-configuring descendant nnU-Net [9] encode a volume through a downsampling encoder into a feature pyramid, then decode back to full resolution via a mirrored upsampling path with skip connections carrying fine detail past the bottleneck. Deeper encoder stages have coarser spatial extent and more channels; shallower stages retain spatial detail at low channel count. This split – a coarse, semantically rich bottleneck feeding a computationally light attention stack, and fine, spatially precise early stages reserved for the decoder – is exactly what the coarse/fine tap split in subsection 3.1 reuses. UNETR [7] is a prior example of combining a convolutional decoder with transformer attention for 3D segmentation.

Attention. Scaled dot-product attention [14] computes, for each query token, a weighted average of value vectors, with weights given by the similarity between the query and each key:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V. \quad (1)$$

Multi-head attention runs several such operations in parallel subspaces and concatenates the results, letting different heads attend to different relationships in the same input.

Position encoding. Attention itself is permutation-invariant, so position must be injected separately. Rotary position embedding (RoPE) [12] does this by rotating each query/key vector by an angle proportional to its position before the dot product, so the result depends only on the *relative* position between the two tokens rather than their absolute indices. In 3D, this is applied axially: the head dimension is split into three chunks, one per spatial axis, each rotated by an angle proportional to that axis’s physical coordinate (scaled by voxel spacing) – letting attention between two cells at different cascade resolutions still reflect their true physical distance, not just their grid index.

2.3 Synthetic Training Data

Domain randomization trains a model on procedurally varied synthetic data so it does not overfit to the narrow appearance distribution of a single acquisition protocol or training cohort. SynthSeg [1] demonstrated this at scale for medical segmentation: training pairs are generated by sampling a random Gaussian-mixture appearance conditioned only on a label map, so the resulting segmentation network never sees a real image of any specific contrast during training yet generalizes across MRI contrasts and resolutions. GIN/IPA [10] takes a complementary, appearance-only approach to single-source domain generalization: a random nonlinear intensity warp (GIN), optionally blended across spatially distinct regions of the same volume (IPA) to break spurious global appearance correlations, achieving strong generalization from randomizing appearance alone, without touching shape or geometry. Both results motivate treating appearance randomization as a distinct lever from geometric augmentation – a question the synthetic task generator (subsection 3.4) revisits directly by ablating *how* painted appearance is randomized (i.i.d. vs. spatially correlated texture noise) while holding geometry fixed.

2.4 Related Work

Three prior systems attempt in-context medical image segmentation, each making a different choice on the three axes this thesis organizes around.

UniverSeg [2] popularized the approach with a CrossBlock architecture that densely cross-attends target and context features at full resolution – no coarse-to-fine structure, and no synthetic-task training.

Iris [6] encodes each context pair into a compact per-class task embedding and fuses it into the target through a pixel shuffle-unshuffle operation – a single fusion step, applied once, with no iterative refinement across scales.

Table 1: Positioning against prior in-context segmentation models on the three axes this thesis organizes around.

	Fusion	Coarse-to-fine	Synthetic tasks
UniverSeg	dense cross-attn.	no	no
Iris	one-shot pixel shuffle	no	no
Medverse	early concatenation	full-volume re-scan	no
Ours	dual attention (repeated)	region-restricted, register-carried	GMM repaint + texture

Medverse [8] performs next-scale autoregressive segmentation, inspired by next-scale image generation [13]: at each step, the target and the previous step’s (perturbed) prediction are concatenated and processed by a shared U-Net, together with the real context pairs, correlated via a blockwise cross-attention module that pools queries and keys to block resolution for efficiency (subsection 3.2). Every step still re-scans the *entire* volume with a sliding window – only the input changes with scale, not the spatial extent processed.

Deep Neural Patchworks [11] stacks levels of increasing resolution for large-volume segmentation, but targets single-model supervised segmentation rather than in-context, few-shot prediction – an architectural precedent for coarse-to-fine cascades outside the ICL setting.

3 Methodology

3.1 Problem Setup and Backbone

Given a target volume I^t and a context set $\{(I_k^c, L_k^c)\}_{k=1}^K$ of image/mask pairs for the same anatomical class as the (unknown) target mask, the model predicts a target mask $\hat{L}^t$. Prediction proceeds through a coarse-to-fine cascade of M resolution levels with decreasing voxel spacing $r_1 > r_2 > \dots > r_M$; each level produces a full mask that seeds the next, finer level (subsection 3.3). Figure 1 shows our architecture (Medverse’s, for comparison, is shown in Figure 4 in the appendix).

Shared encoder. All $K + 1$ volumes of a task are passed once through a shared encoder, producing a feature pyramid per volume. Two disjoint subsets of its stages are read out: *coarse* stages, resampled to a common $R \times R \times R$ token grid ($N = R^3$ cells per volume) and consumed by the attention stack below; and *fine* stages, kept at native resolution and reserved for the decoder’s skip connections (subsection 3.2). This split keeps the attention stack on a small, fixed-size token grid while still allowing a full-resolution decoded mask.

Tokenization. Each cell of every volume yields two tokens sharing one row – an image token z^{img} and a label token z^{lbl} – assembled into a single sequence together with a small number of learned *register tokens*, shared across all volumes, that give the attention stack fixed-size read/write capacity not tied to any single spatial cell. Context rows carry the ground-truth mask in their label token; the target’s does not, since that is exactly what is being predicted – it instead starts from a neutral placeholder or, at cascade levels beyond the first, the previous level’s own prediction (subsection 3.3).

Draft diagram, ported from the conference draft; finalize labels/scale once the architecture is settled.

3.2 Dual Attention: Modality and Position Axes

How a model combines image and label evidence, and how context and target interact, are two related design choices this section addresses together. Medverse [8] concatenates image and label channels before a shared U-Net; context and target then interact through a Blockwise Cross-Attention Module (BAM) placed at two points in that network (once encoder-side, once decoder-side), which pools its queries and keys to block resolution before attending – trading positional precision for a cheaper attention cost – while its values stay at full resolution. Iris [6] fuses image and label features once, through a pixel shuffle–unshuffle operation, then lets context inform the target through a single bidirectional cross-attention step against a compressed, per-class task embedding computed once and reused across every query. Both commit every design choice – fusion point and context–target interaction alike – to a single, early step, computed once.

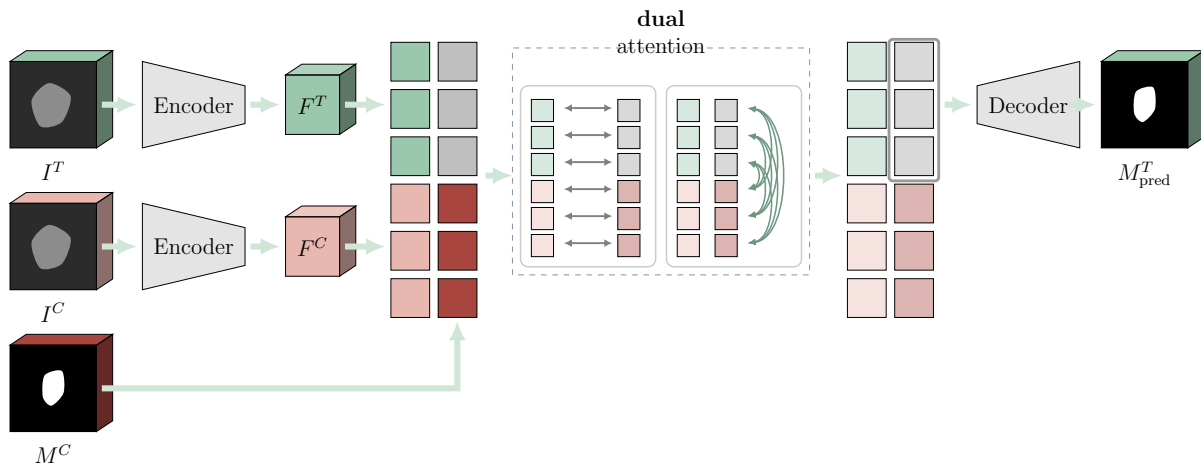


Figure 1: Architecture overview: separate image/label streams + dual attention.

We instead keep image tokens z^{img} and label tokens z^{lbl} (subsection 3.1) as separate streams throughout the network and let them exchange information repeatedly, through *dual attention* – following the general pattern of Fu et al. [5], who fuse a position-wise and a channel-wise attention branch at every layer of a scene-segmentation network – with two attention axes applied at every layer, on the full, uncompressed token grid:

- **Modality axis:** at each cell, the image and label token attend to each other, letting label evidence sharpen the image representation and vice versa, independently per cell.
- **Position axis:** every cell of every volume – context, target, and the register tokens – attends over the entire token sequence at once, using 3D axial rotary position embeddings so attention is a function of physical distance between cells rather than raw grid index. This single joint attention performs within-volume self-attention and cross-context matching together, since tokens from every volume already share one sequence.

Figure 2 illustrates both axes within a single layer.

Only the target's post-attention *image* token is read out for decoding; the label token is discarded after the last layer, since it started from a placeholder rather than real image evidence for the target row. The decoded image tokens are projected back to the coarse token grid and progressively fused with the encoder's fine-resolution skip features (subsection 3.1) via upsampling and convolution, one step per fine stage, into the final full-resolution mask logits.

Draft diagram, ported from the conference draft; finalize once the layer factorization

3.3 Coarse-to-Fine Cascade

The cascade connects consecutive resolution levels through three mechanisms, each replacing a coarser design choice made by Medverse or Iris.

Query prior. Medverse [8] feeds an upsampled, perturbed ground-truth mask into an extra U-Net branch at every step; Iris [6] uses no prior at all. At level $\ell > 1$, we instead initialize the target’s label token directly from level $\ell-1$ ’s own prediction:

$$z_{\ell,0}^{\text{lbl}} = \text{LblEmbed}(\text{Warp}_{\ell-1 \rightarrow \ell}(\hat{L}_{\ell-1}^t)), \quad (2)$$

where Warp resamples level $\ell-1$ ’s prediction onto level ℓ ’s crop grid, correcting for the two levels’ different centers, field of view, and any augmentation applied independently to each. This makes the previous level’s belief part of the same token stream processed by dual attention alongside the image (subsection 3.2), rather than a separate side channel. Figure 3 shows this flow across two levels.

Region-restricted fine processing. Medverse and Iris re-scan the entire volume with a sliding window at every step. Once a coarse level has localized the structure of interest, we instead restrict the sliding window at the next, finer level to the coarse prediction’s foreground region, thresholded and padded by a fixed margin. Compute at fine levels then scales with the size of the structure rather than the size of the input volume, at the cost of depending on the coarse level’s localization being approximately correct.

Cross-level training via register tokens. Medverse and Iris connect levels only through the rendered prediction feeding the next step, with no gradient path between them. The register tokens already introduced in subsection 3.1 give this a second, learned path: their final state $\mathbf{r}_\ell$ at level ℓ is carried into level $\ell+1$ via a learned projection and a level-independent type embedding:

$$\mathbf{r}_{\ell+1,0} = W_r \mathbf{r}_{\ell,\text{final}} + \mathbf{e}_{\ell+1}. \quad (3)$$

Unlike the query prior, this path is not detached between levels: level $\ell+1$ ’s loss can backpropagate into level ℓ ’s own attention, giving the model a direct incentive to carry forward whatever helps the next level – e.g. uncertainty or cues about neighboring structures – beyond what the rendered mask alone can encode.

Draft diagram, ported from the conference draft; finalize once the number of levels M is settled.

3.4 Synthetic Task Generation

Real (image, mask) pairs are limited to the anatomical classes and appearance distribution covered by the training cohort. To broaden training beyond this, a fraction p_{synth} of training tasks are drawn from a synthetic generator instead of real context/target subjects.

Supervoxel repainting. A real-anatomy volume is first oversegmented into supervoxels (unsupervised, appearance-agnostic regions). One supervoxel is treated as the target class, and its binary mask is label-perfect by construction. Each of the $K+1$ volumes needed for a task is then produced by repainting the same supervoxel geometry with independently sampled per-class Gaussian intensities (a Gaussian mixture over classes), giving $K+1$ (image, mask) pairs that share identical anatomy but differ in appearance – no real cross-subject registration or annotation is required.

Paint noise. The default painting noise is i.i.d. per voxel: it matches each class’s target intensity variance but has near-zero spatial autocorrelation, unlike real CT/MRI tissue (measured ≈ 0.6 autocorrelation at native resolution, versus ≈ 0.0 for i.i.d. paint). An optional multi-octave variant instead sums several coarse-to-fine random fields before painting, producing spatially correlated texture that better matches real tissue statistics while leaving the per-class intensity variance unchanged.

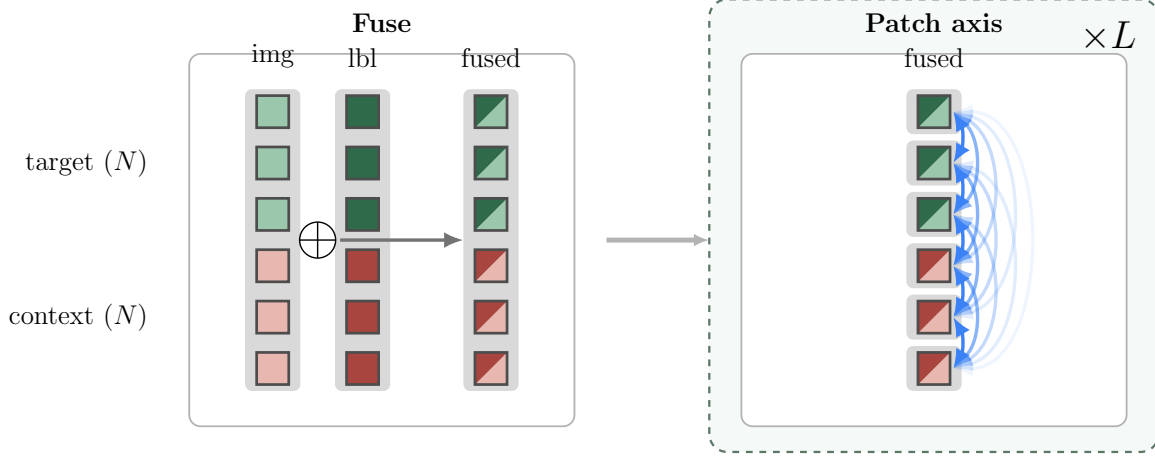
3.5 Training Objective

Each cascade level is supervised independently at its own resolution with a combined binary cross-entropy and soft Dice loss, summed over the M levels:

$$\mathcal{L} = \sum_{\ell=1}^M \lambda_{\ell} (\mathcal{L}_{\text{BCE}}^{\ell} + \mathcal{L}_{\text{Dice}}^{\ell}) . \quad (4)$$

Level weights λ_{ℓ} are uniform ($\lambda_{\ell}=1$) in the reference configuration; the query-prior path is detached between levels (subsection 3.3), so each level’s loss trains its own prediction without back-propagating through the coarser level’s mask.

Baseline: early fusion



Ours: dual attn.

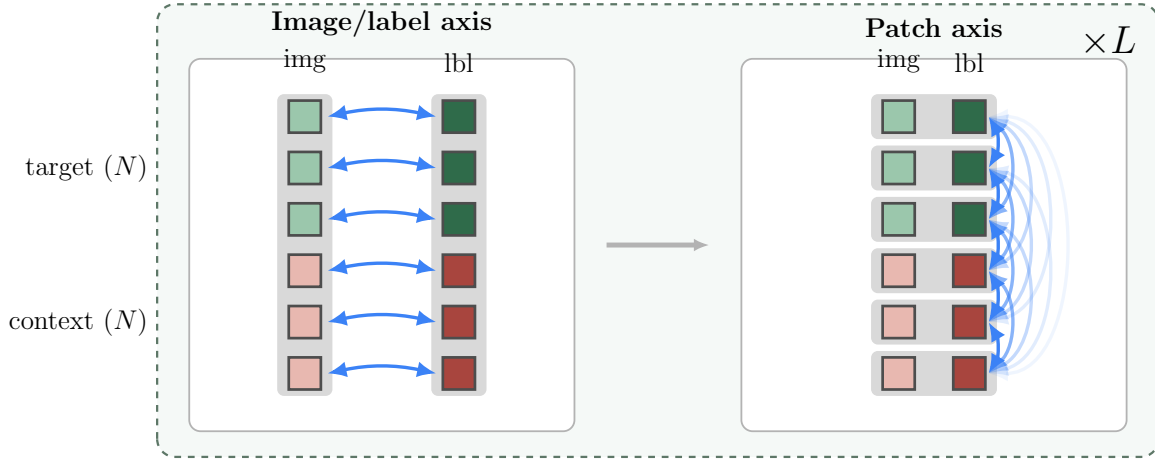


Figure 2: Dual attention within one transformer layer, shown for a single context volume ($K=1$ for clarity) and 3 of its N spatial tokens per stream. Left: the modality axis – image and label tokens attend to each other within the same cell. Right: the position axis – tokens attend across cells and, for the target, across contexts. Green edges touch the target row, which cross-attends read-only into context and whose label-stream input is the query prior (subsection 3.3) rather than ground truth.

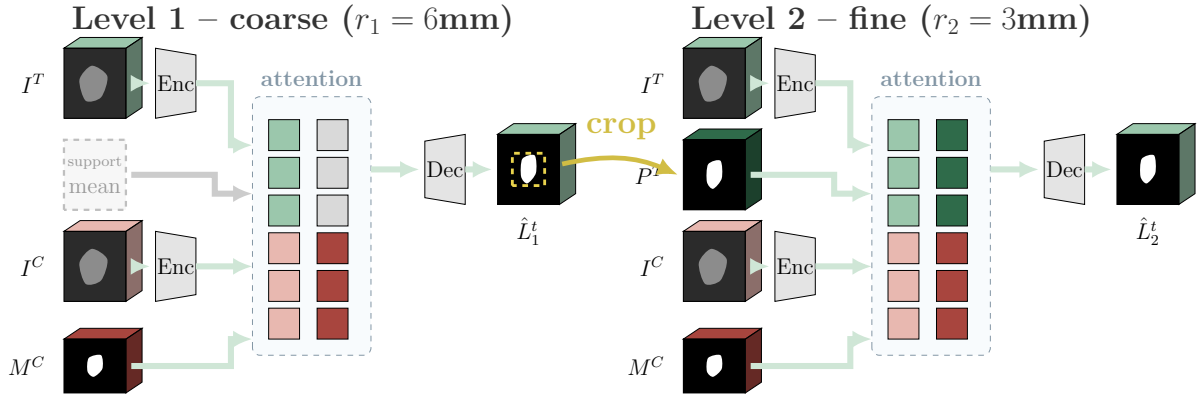


Figure 3: Query image prior across two cascade levels, with the full model input/output flow. At level 1 the target has no real prior yet (dashed support-mean placeholder); at level 2 the target's second lane is a real crop of level 1's own prediction $\hat{L}_1^t$ – the same lane the context mask always occupies.

4 Results

4.1 Experimental Setup

Model scope. Two configurations of the architecture are reported in this chapter, kept explicitly separate. The reference configuration (dual attention, query-prior init, register carry gated off) is the source of every Fusion and Cascade number below. An extended configuration (adding cascade registers, a pooling token, and synthetic texture noise on top of it) is the source of every Synthetic Task Generation number. The two are trained independently and not directly comparable to each other – the extended configuration is read as a promising newer result, not a replacement for the numbers reported above it.

Baselines. Medverse [8] (released weights, and – where noted – finetuned on the same TotalSegmentator split) is the only baseline evaluated empirically here.

Metrics. Accuracy is reported as Dice (macro, per-class then averaged, unless stated per-sample) and Normalized Surface Dice (NSD). Compute is reported as GFLOPs, wall-clock latency, and peak VRAM where available; VRAM comes from a standalone $B=1, K=1$ inference benchmark rather than the evaluation loop itself, so it should not be read as representative of a real $K>1$ eval run.

Datasets. Table 2 lists every evaluation source used in this thesis, grouped by relationship to the training vocabulary. TotalSegmentator (the in-distribution source) and FLARE22 both reuse the training vocabulary [15], splitting into a genuine seen/unseen pair; the rest introduce a target, class, or scanner modality absent from training entirely.

*Sourced from an unofficial data mirror; provenance not yet independently confirmed.

4.2 Image–Label Fusion

Two comparisons follow, each isolating one design choice. The first asks whether moving from Medverse’s dual U-Net and cross-attention to a shared encoder with attention already helps, before anything else about this thesis’s architecture is considered; the second asks whether keeping image and label evidence as separate streams – dual attention – helps further, within that same encoder-and-attention family.

Setup. Table 1 classifies Medverse’s fusion strategy as early concatenation: image and label channels are merged before a shared U-Net processes them, committing to a single fusion point at the very start of the network. This arm is labeled **Early Concat** throughout, not Medverse: it starts from Medverse’s released weights (the best

Resolve before submission: does the thesis report a single “Ours” configuration, or keep this split?

No Iris baseline exists yet; the comparison in subsection 2.4 remains architectural, not empirical.

Add citations for each source

Table 2: Every evaluation source, grouped by relationship to the training vocabulary. n is subjects; n_{cls} is evaluated classes.

Source	Modality	Target	n
<i>In-vocabulary (seen/unseen split)</i>			
TotalSegmentator	CT	117 organs (61 seen / 56 unseen)	89
FLARE22	CT	abdominal organs (11 seen / 2 unseen)	50
<i>Novel vocabulary (unseen by construction)</i>			
ISLES22	MRI DWI	stroke lesion	250
Shifts-MS	MRI FLAIR	MS lesion	46
MSD Hippocampus	MRI T1	hippocampus	260
MSD Prostate	MRI T2+ADC	prostate zone	64 (32 cases $\times$ 2 channels)
ATLAS v2.0*	MRI T1	chronic stroke lesion	654
GNC_705	MRI Dixon	kidney lesion	610
NasalSeg	CT	nasal/sinus cavities	107
<i>Non-object</i>			
HU_LWK1	CT	L1 vertebra ROI	36
<i>Held-out modality</i>			
TotalSeg MRI	MRI	same classes as TotalSegmentator CT	47

available starting point for this architecture) but is then trained under the identical recipe as the other two arms for the same ~ 250 epochs – a different thing from the released, unmodified baseline evaluated elsewhere in this thesis (subsection 4.3). The single-axis control isolates the architectural family this comparison targets: it keeps the shared encoder and attention stack common to every architecture in this thesis, but collapses image and label evidence into one token per cell via a pixel shuffle–unshuffle operation *before* the transformer – echoing Iris’s own fusion mechanism in spirit though not in implementation, and withholding dual attention until the next step of the staircase.

Its transformer depth is raised from $l=4$ to $l=9$ specifically to compute-match the dual-attention arm (removing the modality axis also removes a large share of the model’s compute; parameter count and compute could not be matched simultaneously, leaving parameters 52% higher for the single-axis control). This adjustment is made relative to dual attention only, not to Early Concat, so the Early Concat comparison alone is not compute-matched.

All three arms train under an identical recipe (Table 3), an identical data pipeline, and an identical augmentation recipe, but not an identical starting point: Early Concat keeps its released initialization, while the single-axis and dual-attention arms warm-start only the encoder and decoder from a shared checkpoint, with the entire in-context reasoning core (transformer, image/label embeddings, context/query id, register tokens, cascade

Table 3: Training settings shared by Early Concat, the single-axis control, and dual attention (ours). Epoch count is a placeholder pending confirmation of the exact per-arm value.

Optimizer	AdamW
Learning rate	10^{-4}
Schedule	cosine decay, 10-epoch warmup
Weight decay	10^{-2}
Batch size	8
Gradient clipping	1.0
Loss	BCE + Dice
Epochs	~ 250

projection, pool projection) randomly initialized for both – so neither patchset arm starts from an already-trained transformer. Training uses real anatomy only ($p_{\text{synth}}=0$), CT only. The three arms also did not train to a matched stopping point: Early Concat trained to full convergence, the patchset arms did not.

All three arms are then evaluated with an identical protocol – fixed 3 mm single-level crops, fp32, each model’s own best checkpoint – across every dataset this thesis evaluates (subsection 4.1), not TotalSegmentator alone.

Rows below are grouped by each dataset’s relationship to the training vocabulary, not by how different the source otherwise looks: TotalSegmentator itself (61 seen / 56 unseen classes) and FLARE22 (11 seen / 2 unseen, out of 13 total, subsection 4.1) both reuse the training vocabulary and so split into a genuine seen/unseen pair; MSD Prostate and MSD Hippocampus introduce organs with no vocabulary overlap at all, so every class in that pair is unseen by construction, not by a train/val split; HU_LWK1’s target is a HU-measurement region, not a segmentable anatomical object, a categorically different kind of “unseen” from a novel organ; TotalSegmentator’s own MRI acquisition isolates a held-out *modality* instead, over classes the vocabulary already contains.

Results. Single-axis alone already beats Early Concat on five of the eight rows – both TotalSegmentator splits, both FLARE22 splits, and TotalSeg MRI – while Early Concat wins the three remaining rows: MSD Prostate, MSD Hippocampus, and HU_LWK1. Restoring dual attention flips one of those three (MSD Hippocampus, 0.184 vs. 0.172) without touching the other two, bringing the full architecture to six wins out of eight against Early Concat. The one aggregate number in the table – mean of the five held-out sources – shows why that flip matters: single-axis alone is a net regression against Early Concat there (0.264 vs. 0.278), and only once dual attention restores the modality axis does the full architecture overtake Early Concat (0.297). The architectural family change alone, in other words, is not what carries this result; dual attention is. The

Table 4: Early Concat, the single-axis control, and dual attention (ours), across all evaluation datasets (subsection 4.1). Grouped by vocabulary relationship: in-vocabulary seen/unseen, novel-vocabulary (unseen by construction), non-object, held-out modality.

	Early Concat	Encoder + Attention	
		Single-axis	Dual attention (ours)
TotalSeg CT – seen	0.348	0.574	0.582
TotalSeg CT – unseen	0.221	0.408	0.426
FLARE22 – seen	0.532	0.693	0.709
FLARE22 – unseen	0.443	0.632	0.604
MSD Prostate (unseen)	0.377	0.205	0.247
MSD Hippocampus (unseen)	0.172	0.122	0.184
HU_LWK1 (non-object)	0.146	0.079	0.102
TotalSeg MRI (modality)	0.178	0.232	0.259
mean of 5 held-out sources	0.278	0.264	0.297
l	–	9	4
params	71.1M	74.1M	48.7M
GFLOPs	2362.6	3973.1	3892.5
latency (fp32, per sample)	73.2 ms	73.0 ms	72.5 ms

compute picture is mixed rather than uniformly favorable: dual attention has the fewest parameters and the lowest latency of the three, but not the fewest GFLOPs – Early Concat remains cheapest by that measure despite its lower accuracy, so no arm here is a strict win on both accuracy and compute at once. Two caveats keep this from a single-variable ablation: Early Concat starts from a pretrained initialization rather than the randomly-initialized reasoning core the patchset arms share, and does not receive the same query-conditioning signal during training – so this is best read as a comparison against Early Concat’s strongest available starting point, not a fully symmetric ablation.

Status. Neither patchset arm had reached its full training budget at the point these numbers were read, while Early Concat had already converged – so Table 4 may understate Early Concat’s eventual ceiling more than either patchset arm’s. All three arms are N=1 seed, and several rows rest on very few classes (FLARE22’s unseen split, both MSD sources, HU_LWK1) – directional evidence, not settled effect sizes.

4.3 Coarse-to-Fine Cascade

This section asks whether the coarse-to-fine cascade introduced in subsection 3.3 helps in-distribution at all; how the resulting design compares to Medverse’s own coarse-to-fine mechanism, on accuracy and compute; and which of the cascade’s own mechanisms are responsible for the in-distribution gain.

Re-read once
Early Concat receives the same query-conditioning signal – currently dropped due to a training instability under

Table 5: Predicted-prior checkpoint vs. Medverse (native autoregressive, depth-matched), macro Dice and mean per-sample latency. Levels/spacing is our own cascade ladder for that dataset; Medverse’s AR depth is matched to the same level count.

Dataset	Levels	Spacing (mm)	Dice		Latency	
			Ours	Medverse	Ours	Medverse
TotalSeg CT	3	[6, 3, 1.5]	0.586	0.103	278ms	4978ms
FLARE22	3	[6, 3, 2.5]	0.751	0.359	311ms	4894ms
MSD Prostate	3	[3, 1.5, 0.75]	0.375	0.357	279ms	5006ms
MSD Hippocampus	1	[0.5]	0.303	0.686	99ms	50ms
HU_LWK1	3	[6, 3, 1]	0.152	0.020	439ms	5336ms
TotalSeg MRI	3	[6, 3, 1.5]	0.506	0.038	306ms	4842ms
NasalSeg	2	[1.5, 0.8]	0.547	0.737	161ms	1886ms
mean			–	–	268ms	3856ms

Coarse-to-fine cascading improves accuracy in-distribution. A spacing sweep from 4mm to 1.5mm crop resolution (117 classes, 3897 samples) raises macro Dice from **0.361 to 0.538** (+0.177). The locator’s containment of the target structure is already high at the coarse level (0.922 mean, vs. an oracle ceiling of 0.970 – a 0.048 gap) and does not predict which classes gain the most at the fine level: once a class is roughly localized, the remaining error is dominated by segmentation quality at the fine crop, not by further localization failure.

Ours vs. Medverse, seven datasets, accuracy and compute. “Ours” here is the predicted-prior checkpoint (the best-performing configuration of the query-prior ablation below); “Medverse” is released weights, native autoregressive inference – its own coarse-to-fine mechanism – with its refinement depth level-matched to our own cascade ladder per source – the fairer comparison, even though matching that depth actually *hurts* Medverse’s own accuracy. Four held-out sources (ISLES22, Shifts-MS, ATLAS v2.0, GNC_705) are excluded – both models score too low there to be informative. Table 5 reports macro Dice and mean per-sample latency for both.

Ours wins 5/7 (TotalSeg CT, FLARE22, MSD Prostate, HU_LWK1, TotalSeg MRI); Medverse wins 2/7 (MSD Hippocampus, NasalSeg) – both cases where the target is a single, large, well-defined structure that suits Medverse’s full-FOV multi-resolution pyramid. Compute favors Ours by roughly 14× on mean latency – and this is *after* level-matching Medverse’s own depth to ours, which costs Medverse accuracy too, so an unmatched Medverse would lose by even more on both axes at once. Medverse’s own coarse-to-fine mechanism is separately prior-sensitive: feeding its own imperfect coarse prediction forward as a query prior hurts its harness cascade universally, across all 7 OOD sources tested, sometimes catastrophically (MSD Hippocampus 0.691 → 0.050)

TotalSeg CT’s Medverse row is not yet the level-matched run (an earlier, mismatched depth run sub-

Table 6: Query-prior source ablation, macro Dice, across all validation datasets. TotalSeg CT also reports NSD (0.576 / **0.631**).

Dataset	GT-prior (perturbed)	Predicted prior (real prev. pred)
TotalSeg CT	0.543	0.586
FLARE22	0.731	0.751
MSD Prostate	0.310	0.375
MSD Hippocampus	0.319	0.303
HU_LWK1	0.099	0.152
TotalSeg MRI	0.449	0.506
NasalSeg	0.399	0.547

– compounding coarse-level error appears to be a property generic to coarse-to-fine cascades, not specific to either architecture’s design.

A real predicted prior beats a perturbed ground-truth prior. Same checkpoint chain, same seven validation datasets as Table 5 plus in-distribution TotalSeg CT (Table 6):

Feeding the model’s own real previous prediction forward beats a perturbed ground-truth prior on 99 of 117 classes (median $\Delta + 0.043$) – a *prior-source* ablation (real vs. noisy synthetic), distinct from the Medverse *prior-presence* finding above; a true no-prior arm is still needed (gap G6). Out-of-distribution (9 held-out sources including FLARE22/NasalSeg): the prior-source gain helps 5/9 sources and hurts 4/9 – weaker than the clear in-distribution majority.

4.4 Synthetic Task Generation

This section isolates the effect of the synthetic generator itself (subsection 3.4) – paint noise texture and shape/intensity calibration – from the model’s training data mix. All runs below are trained on CT only; the separate question of what adding MRI training data does to generalization is a training-regime choice, not a property of the synthetic generator, and is reported elsewhere rather than folded into these numbers.

Texture noise beats flat noise beats real-only. A controlled three-arm chain shares one starting checkpoint and varies only the paint-noise mechanism: real-only ($p_{\text{synth}}=0$), synthetic with i.i.d. paint noise ($p_{\text{synth}}=0.3$), and synthetic with multi-octave correlated texture noise (same p_{synth}). Metric: TotalSegmentator validation Dice, seen and unseen class macro splits, at a shared epoch-50 cutoff.

Texture noise wins on all three metrics – notably it recovers the seen-class cost that flat

Add a no-query-prior control arm to complete the ablation table promised in the methodology chapter (full model vs. –query-prior vs.

Table 7: Paint-noise ablation, TotalSegmentator validation Dice, epoch 50. Neither synthetic arm had converged at this cutoff (both still rising); the real-only arm had already plateaued.

Arm	Dice	seen	unseen
Real-only	0.4868	0.5737	0.3886
Flat i.i.d. noise	0.4899	0.5676	0.4022
Multi-octave texture	0.4955	0.5748	0.4059

Table 8: Measured target-vs-surrounding-tissue contrast, used to set the shape generator’s calibration range. Sign and magnitude match each source’s known imaging contrast pattern.

Source	Known pattern	Contrast (ring- σ)
ISLES22 (DWI, acute stroke)	hyperintense	+2.42
Shifts-MS (FLAIR, MS lesion)	hyperintense	+1.49
GNC_705 (kidney lesion)	mixed cyst/complex	+0.49
MSD Prostate (zones)	–	+0.35
HU_LWK1 (vertebra ROI, CT)	≈ 0 expected	–0.07
MSD Hippocampus	–	–0.51
ATLAS v2.0 (T1, chronic stroke)	hypointense	–0.59

noise alone pays (narrowly beating even the real-only arm), while matching flat noise’s unseen-class gain.

Shape and intensity realism, calibrated against measurement. The procedural shape-mode generator’s host-anchored intensity contrast (subsection 3.4) was calibrated against a fresh measurement of real target-vs-surrounding-tissue contrast, taken directly on the same held-out sources used for generalization testing (749 cases, effect size in ring-standard-deviation units) rather than guessed.

Every source’s sign and rough magnitude matches its known radiological contrast direction, validating the measurement before it was used to set the generator’s calibration range. The measurement also caught a design error: an initial *absolute*-offset version of this calibration was, by this same measurement, roughly 7–12 $\times$ too extreme relative to the generator’s own per-class variance scale – corrected to a ratio-of-host-variance design before any reported checkpoint used it.

Not yet isolated from the training-mix change. Two further generator changes – cross-class intensity/variance correlation and widened shape diversity – exist and have been evaluated, but only on top of a checkpoint that concurrently switched the training data mix from CT-only to CT+MRI. Their standalone effect, isolated from that mix change, has not been measured.

5 Discussion

6 Conclusion

ToDoS

Select the correct language in the first rows of `main.tex`. The % makes a line a comment.

Generate a PDF (Visual Studio Code: click on the green arrow in the top right corner).

Check the PDF (Visual Studio Code: click on the two rectangles with the magnifying glass in the top right corner).

Talk about this template with your supervisor: Should you add a declaration for the use of AI? Currently no official AI-declaration is given. You can find one in the pull requests of the github repository.

With CTRL pressed you can click on the PDF and jump to the corresponding place in your `.tex` files.

Check the Errors. (Visual Studio Code: bottom left) There should be one warning because the first ToDo pushes down the second one: LaTeX Marginpar on page 2 moved.

In the `main.tex` you can press ALT + SHIFT and the ARROW UP and DOWN keys to create a multiline cursor, comment out all the sections you are currently not working on and remove the multiline cursor with ESC.

Keep the following examples and todos till the end and check them out sometimes.

Adjust your License Notes to reflect the used programs.

For Printing: Remove the `todonotes`-package in `setup.tex` to find all Todos.

Table 9: A nice table with a nice caption.

Value & Unit	Value	Unit
9.82 m s ⁻²	9.82	m s ⁻²

For Printing: Choose the correct formatting (search `print` in `setup.tex`).

You can remove the glossary and the symbolslist to make it easier, especially in a Bachelor's Thesis.

6.1 Tables and References

Table 9

This is a cited book [3].

Store and display plots and other graphs as `.pdf` files.

Include Algorithm 1 in the same way you include equations like Equation 5:

$$\begin{aligned}y &= 3x^2 + 5x^2 + 4x + 4x + 7 + 1 \\&= 8x^2 + 8x + 8 \\y &= 8 \cdot (x^2 + x + 1)\end{aligned}\tag{5}$$

6.2 Auto-complete in VSCode

There are also some auto-complete features: type `@` + letter to get the letters greek version.

Type `@v` + letter for the alternative form (like φ for ϕ).

You can also use shortcuts for environments like

$$BAL\tag{6}$$

Type BAL and select the correct option from the autocomplete. Maybe you have to try various ones. The B is the base command and AL is the abbreviation for align.

If you want an environment without an index, you can use BS as base command.

Same for changes to your font, like bold or italic: FIT gives *text*

6.3 ", Quotations and Code

Use " or file-names, file_names and code_snippets.

Algorithm 1: The Colors from the Corporate Design of the University of Freiburg. The keywords are bold and the comments blue. This can be changed in setup.tex.

```
from dataclasses import dataclass
```

```
# https://cd.uni-freiburg.de/farben/
```

```
@dataclass
```

```
class MainColors:
```

```
    blue = "#344A9A"
```

```
    black = "#000000"
```

```
    white = "#FFFFFF"
```

```
@dataclass
```

```
class BackgroundColors:
```

```
    blue_100_percent = "#344A9A"
```

```
    blue_80_percent = "#5D6BAD"
```

```
    blue_60_percent = "#868DC2"
```

```
    blue_80_percent = "#AFB1D8"
```

```
    blue_20_percent = "#D7D8EC"
```

```
    dark_blue = "#00004a"
```

```
    sand_100_percent = "#f6f1e3"
```

```
    sand_50_percent = "#faf8f1"
```

```
@dataclass
```

```
class AdditionalColors:
```

```
    green_100_percent = "#00a082"
```

```
    green_80_percent = "#27b29b"
```

```
    green_60_percent = "#7bc6b4"
```

```
green_80_percent = "#afdace"  
green_20_percent = "#daede7"  
brown_100_percent = "#8f6b30"  
brown_80_percent = "#a58959"  
brown_60_percent = "#bca683"  
brown_80_percent = "#d2c4ac"  
brown_20_percent = "#e9e1d6"  
yellow_100_percent = "#ffe863"  
yellow_80_percent = "#ffed82"  
yellow_60_percent = "#fff1a1"  
yellow_80_percent = "#fff6c1"  
yellow_20_percent = "#fffae0"  
pink_100_percent = "#f5c2ed"  
pink_80_percent = "#f7cef1"  
pink_60_percent = "#f9daf4"  
pink_80_percent = "#fbe7f8"  
pink_20_percent = "#fdf3fb"
```

6.4 Clipboard Manager, Git and more

Clipboard Manager help with copying multiple different things at once. On Windows you can press WINDOWS + V to activate the history and choose older things you copied.

Use Git ([click here](#)).

6.5 Glossary

Click them, the page numbers in the glossaries are also clickable.

name one

name one

name one

long form of name two (short2)

short2

short2

a symbol

a symbol

a symbol

Big Letter at start: Name one

plural of name one

name one

Big Letter at start: Short2

short2

short2s

Big Letter at start: *A symbol*

a symbol

Symbols

Notation	Description	Page List
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List of Algorithms

- 1 The Colors from the Corporate Design of the University of Freiburg. The keywords are bold and the comments blue. This can be changed in `setup.tex`. 22

Glossary

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name one	description one	21, 23
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References

- [1] Benjamin Billot, Douglas N. Greve, Oula Puonti, Axel Thielscher, Koen Van Leemput, Bruce Fischl, Adrian V. Dalca, Juan Eugenio Iglesias, et al. SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining. *Medical Image Analysis*, 86:102789, 2023.
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Appendix

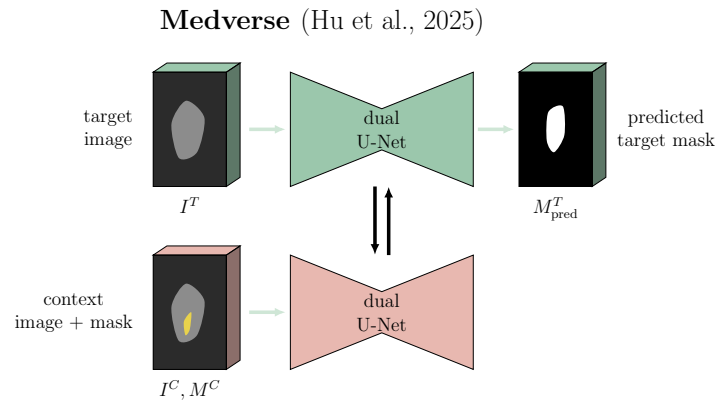


Figure 4: Medverse’s architecture: early concatenation + dual U-Net, shown for comparison against our architecture (Figure 1).